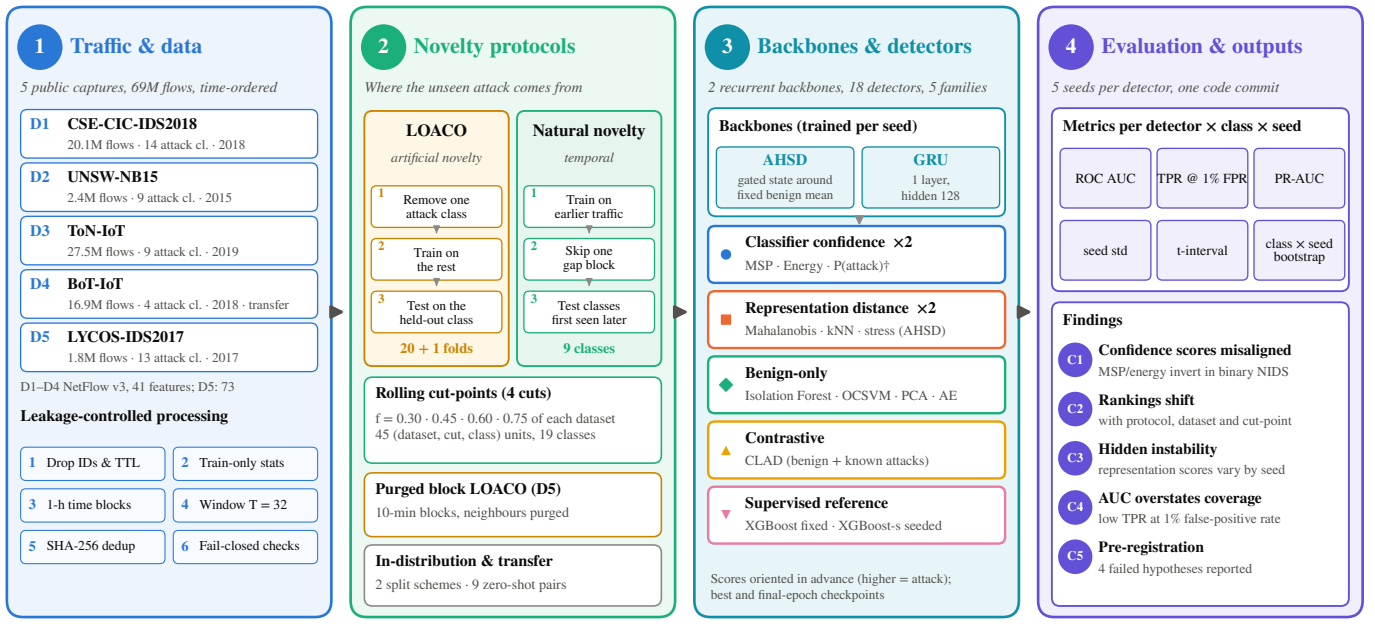
1
Traffic & data
5 public captures, 69M flows, time-ordered
D1 CSE-CIC-IDS2018
20.1M flows · 14 attack cl. · 2018
D2 UNSW-NB15
2.4M flows · 9 attack cl. · 2015
D3 ToN-IoT
27.5M flows · 9 attack cl. · 2019
D4 BoT-IoT
16.9M flows · 4 attack cl. · 2018 · transfer
D5 LYCOS-IDS2017
1.8M flows · 13 attack cl. · 2017
D1–D4 NetFlow v3, 41 features; D5: 73
Leakage-controlled processing
1 Drop IDs & TTL
2 Train-only stats
3 1-h time blocks
4 Window T = 32
5 SHA-256 dedup
6 Fail-closed checks
2
Novelty protocols
Where the unseen attack comes from
LOACO
artificial novelty
1 Remove one attack class
2 Train on the rest
3 Test on the held-out class
20 + 1 folds
Natural novelty
temporal
1 Train on earlier traffic
2 Skip one gap block
3 Test classes first seen later
9 classes
Rolling cut-points (4 cuts)
f = 0.30 · 0.45 · 0.60 · 0.75 of each dataset
45 (dataset, cut, class) units, 19 classes
Purged block LOACO (D5)
10-min blocks, neighbours purged
In-distribution & transfer
2 split schemes · 9 zero-shot pairs
3
Backbones & detectors
2 recurrent backbones, 18 detectors, 5 families
Backbones (trained per seed)
AHSD
gated state around fixed benign mean
GRU
1 layer, hidden 128
Classifier confidence ×2
MSP · Energy · P(attack)†
Representation distance ×2
Mahalanobis · kNN · stress (AHSD)
Benign-only
Isolation Forest · OCSVM · PCA · AE
Contrastive
CLAD (benign + known attacks)
Supervised reference
XGBoost fixed · XGBoost-s seeded
Scores oriented in advance (higher = attack);
best and final-epoch checkpoints
4
Evaluation & outputs
5 seeds per detector, one code commit
Metrics per detector × class × seed
ROC AUC
TPR @ 1% FPR
PR-AUC
seed std
t-interval
class × seed bootstrap
Findings
C1 Confidence scores misaligned
MSP/energy invert in binary NIDS
C2 Rankings shift
with protocol, dataset and cut-point
C3 Hidden instability
representation scores vary by seed
C4 AUC overstates coverage
low TPR at 1% false-positive rate
C5 Pre-registration
4 failed hypotheses reported